



A 17 MS/s SAR ADC with energy-efficient switching strategy

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Abstract

A simple energy-efficient switching procedure is proposed to reduce the total number of switches and facilitate the capacitors matching requirements in SAR ADCs. The main idea is that the coupling capacitor (C_C) is utilized as the same as the unit one, which is applied for LSB charge production. After the reset phase, only the MSB capacitor is connected to the input potential, while the other experience either low and high reference potential; hence, a simplified switching strategy with 55% reduction in number of switches is retrieved. Post-layout simulation results confirm SNDR and ENOB of around 80 dB and 13 bit, respectively, at Nyquist input frequency when the conversion rate is 17 MS/s. In these conditions, FOMW and FOMS reach to 139.9 fJ/Conv-Step and 166.15 dB, respectively. Also, INL and DNL would experience the maximum level of $+0.95/-0.58$ LSB and $+0.63/-0.74$ LSB, respectively. The proposed ADC occupies active area of 1.7 mm^2 and consumes around 20.6 mW power at 5 V supply and 17 MS/s conversion rate. Post-layout simulation results are performed using the HSPICE BSIM3 model of a $0.50 \text{ }\mu\text{m}$ CMOS process.

Keywords ADC · Low power · DAC · Comparator · SAR ADC · High-resolution

1 Introduction

Charge redistribution successive approximation register (SAR) ADCs are commonly demanded for low-power data converter designs, [1–6]. In comparison to the pipeline data converters, power consumption is considerably smaller thanks to eliminating the interstage residue amplifiers between the pipeline stages. Also, the simplicity of downscaling and compatibility to the newer technology nodes are the other benefit of the SAR structure which makes it suitable for low-voltage medium- or high-

resolution Nyquist rate ADC architectures [4, 7–10]. Against the pipeline structures, in which a highly linearized amplifier is required for residue amplification, the SAR ADCs no longer transfers the signal as linear as the input, [10–12]. In pipeline structures, as the supply voltage is reduced, the required headroom remains unchanged and the reduction in available signal swing would degrade SNR, [13–17].

Recently, switched-capacitor SAR ADCs are applied for high resolution applications. In order to achieve higher resolution, a large capacitance is used which suppresses the distortion due to the capacitors mismatch [4, 14, 18–20]. Usually, absolute value of the capacitors are enlarged inside the array structure to minimize the mismatch and other nonidealities; it means a great area and power penalty due to the need to the buffer with high driving capability, [7, 15, 21, 22]. Fundamentally, in conventional capacitive SAR ADCs the total capacitance increases exponentially in relation to the number of bits, while the unit capacitance is limited by thermal noise [1, 9, 13, 18, 23–27]. On the other hand, utilizing a smaller unit capacitance for charge redistribution (CR) digital to analog converters, power and area is decreased. This is known as a trade-off between the linearity and area/power cost in SAR ADCs.

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Several low-power methods have been proposed for resolution enhancement in small chip area, which all emphasis on modifications in capacitive DAC, [4, 5, 18, 28–33]. The key motivation for utilizing SAR-ADCs in such applications is the simplicity of their structure due to the small analog circuitry, [8, 15, 25, 29, 30, 34]. A split capacitor DAC is commonly used to reduce the number of unit capacitors. But, the mismatch and the parasitic capacitances associated with the coupling capacitor degrade the linearity of the internal DAC, [10, 14, 16, 17, 31, 35, 36]. Design of high-resolution SAR ADC is subject to two main challenges; firstly, thermal noise in the comparator circuit and secondly, the nonlinearity due to the mismatch between capacitors. The first one restricts the maximum achievable ENOB and presents a tradeoff between area and power due to using low-noise and high-power comparators, [2, 8, 33]. Dynamic and time-domain comparators, [2, 4, 6, 8], have been proposed with digital-intensive ADC architecture to satisfy this requirements. However, mismatch compensation throughout the capacitors of the binary-weighted structure is rarely focused due to the complexity of handling a large number of elements. In order to enhance the linearity of the split-capacitor DAC, the conventional approaches focused on gain calibration in split capacitor-bank DACs, [17, 31, 34, 37], digital-based calibration of parasitic and mismatch [27], and software-based complicated digital background calibration, [20, 38]. However, extra hardware and power is inevitable, [9, 10, 12, 22, 24, 30, 32, 33, 36, 39, 40].

In this paper a novel technique is proposed to enhance the linearity of the capacitor DAC (CDAC) in fully differential SAR ADC with special focus on minimizing the mismatch due to the parasitic-induced capacitances. Using the LSB capacitor, as the same as the coupling one, considerably improves the mismatch nonidealities, initiated from the error gradient in large chip dimensions. Another benefit is that the total number of switches is decreased by about 55%. The switching strategy is modified such that only the MSB capacitor is connected to the input potential and other capacitors only experience the reference voltage levels. The proposed ADC is designed in a 0.5 μm CMOS process, as was applied for some other works, [16], to improve the fabrication cost and time. In this design, only thick gate transistors with a minimum gate length of 0.5 μm were utilized. Meanwhile, even with such an old process, SAR ADCs can still preserve the low-power advantages over the other types of ADCs. The proposed paper is organized as follows. In Sect. 2, the proposed fully differential SAR ADC would be introduced. Section 3 describes the CMOS implementation of the proposed strategy. Post-layout simulation results are presented in Sect. 4, and finally, Sect. 5 concludes the paper.

2 The proposed fully differential SAR ADC

The building blocks of typical fully differential SAR ADC is shown in Fig. 1. The SAR ADC based on capacitor DAC, contains SAR registers, DAC and the analog voltage comparator. Conversion to digital codes is realized via the analog comparator which evaluates the input signal in comparison to the proper reference potential. In order to extract further bits, the comparison process will continue until the entire digital code is generated. Inherently, data conversion is serially accomplished in SAR ADCs and at least ‘n’ clock cycles are required to extract ‘n’ bits, [1, 4, 8, 10, 26, 31, 33, 37, 40–43]. Based on the results of the recent comparison, a powerful binary search algorithm decides that which reference levels should be applied on capacitive DAC for the next bit extraction. Binary search algorithm and scheduling on different clock cycles are delegated to the SAR registers. Besides the matching restrictions occurred by using different coefficient capacitors in CDAC-based SAR ADCs, absolute value of the unit capacitance and number of switches restricts the conversion speed.

Hence, split CDAC technique is represented in which total capacitance is reduced, [1, 14, 17, 18, 27, 30, 32, 39], however, the number of switches is as the same as the conventional one. On the other hand, in the split CDAC technique size of the coupling capacitor is the most significant parameter which should be carefully considered. In order to overcome the mentioned drawbacks, a novel and reliable switched–merged method is presented.

Figure 2 illustrates the proposed fully differential SAR ADC architecture. Consider that V_{cm} , V_{top} and V_{bot} represent the common-mode voltage level, high and low level of the reference voltages, respectively. Input voltages are also denoted by V_{in+} and V_{in-} . As shown, similar to the other conventional split CDAC structures, data conversion is accomplished via two segmented sections for higher and lower 7-bits. Conversion process is also divided into two phases. The first one is allocated to the sampling phase, in which the bottom plates of MSB capacitors are

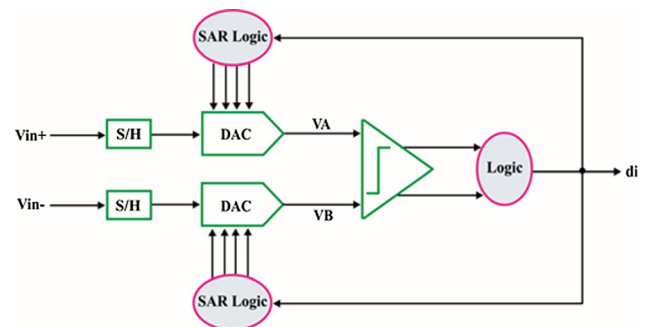


Fig. 1 A general block diagram of the SAR ADC

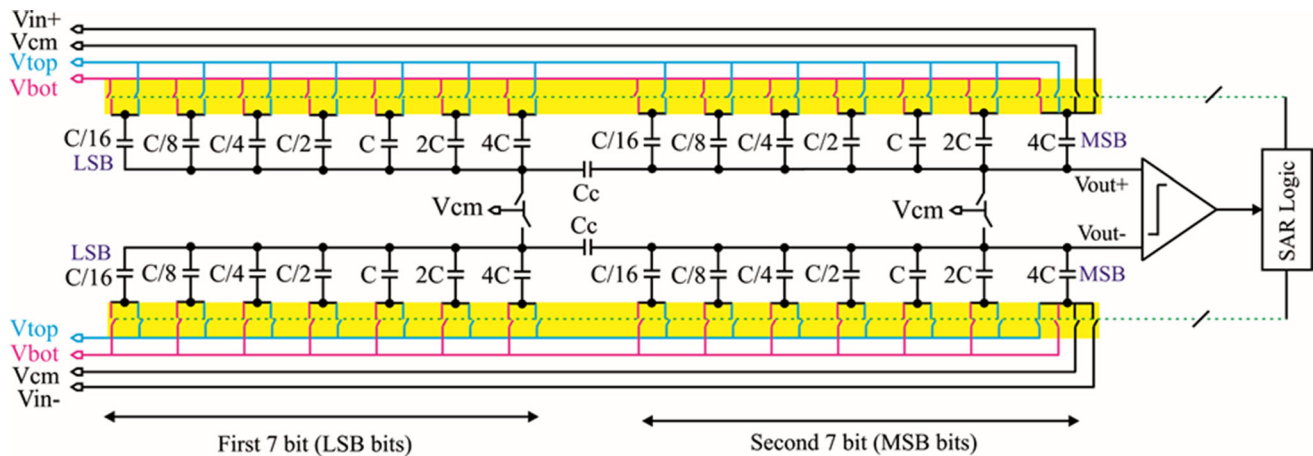


Fig. 2 The proposed fully differential SAR ADC architecture

connected to V_{in+} and V_{in-} , while the rest experience the reference potentials, V_{bot} and V_{top} . During the sampling phase, inputs of the comparator are connected to V_{cm} . Therefore, the MSB capacitors are charged with the input signal. After the sampling phase, in the first conversion cycle the bottom plates of the MSB capacitors would be disconnected from the input signals and connects the V_{cm} . This is for the sake of extracting the first bit which clarifies the sign of the sampled input voltage. In these conditions, inputs of the comparator should be disconnected from V_{cm} . In the next conversion cycles, the bottom plates of the MSB capacitors are connected to V_{cm} , and the rest are connected to V_{top} or V_{bot} , based on the results from the previous decision. Successive approximation conversion flow would continue based on the decision from the last results of the binary search algorithm. As the result, charge distribution would yield the difference of the input signal and the DAC's outputs at the input nodes of the comparator.

Three improvements are achieved thanks to using the proposed technique in comparison to the conventional split capacitor method. Firstly, as it is clear in Fig. 2, total number of switches is decreased, because just the MSB capacitors requires four switches to connect to the input signals, reference voltages or the common-mode voltage level. The rest of the capacitors just require two switches for connecting to the reference voltages. Hence, total number of switches can be compared for the conventional and the proposed strategy, as illustrated in (1) and (2), respectively, in which N represents the number of bits. For instance, number of switches is 116 and 64 number in the conventional and the proposed 14-bit SAR ADC, respectively. This means about 55% improvement.

$$[Total\ switches]_{Conv} = [2(4N + 2)] \quad (1)$$

$$[Total\ switches]_{Prop} = [2(2N + 4)] \quad (2)$$

Secondly, choosing the same coefficient for the coupling and the unit capacitors enhances the linearity by eliminating the error gradient, initiated from the geometrical distribution of the capacitances throughout the great chip area. Thirdly, in the proposed method, due to implementing the same size coupling and LSB capacitors, the parasitic capacitors are eliminated too. A dummy capacitor ring is utilized to provide identical dummy loads on all capacitors. Also the common-centroid technique is used through the layout floorplan to matching enhancement. Figure 3 illustrates the switching sequence for 3-bit examples of the proposed ADC.

In the proposed method, after that the sampling switches are turned off then the comparator directly performs the first comparison with no need to switching on any capacitor. Hence, as depicted in Fig. 3, the proposed switching sequence consumes no energy before the first comparison and operates more effectively, in comparison to the conventional structure. In order to clarify the operation of the proposed ADC, a 3-bit numeral example would be strictly analyzed based on the quantized values.

During the sampling phase, inputs are being sampled on the MSB capacitors at both the positive and negative sides. The rest of the capacitors would connect to V_{bot} and V_{top} reference potentials, while the inputs of the comparator are connected to V_{cm} . Also, during the hold mode, the MSB capacitors are disconnected from the input potentials and are connected to V_{cm} . In this case, the inputs of the comparator are also detached from V_{cm} . The simplified circuits are depicted in Figs. 4 and 5. Based on the charge conservation law, the sampled charges on the capacitors during the sampling mode have to be equal to the charges hold on the capacitors during the hold mode. Equations are presented in (3) and (4), in which the value of the first bit is obtained equal to “1”. Meanwhile, through the bit cycling action, the comparator compares V_A and V_B . If V_A is

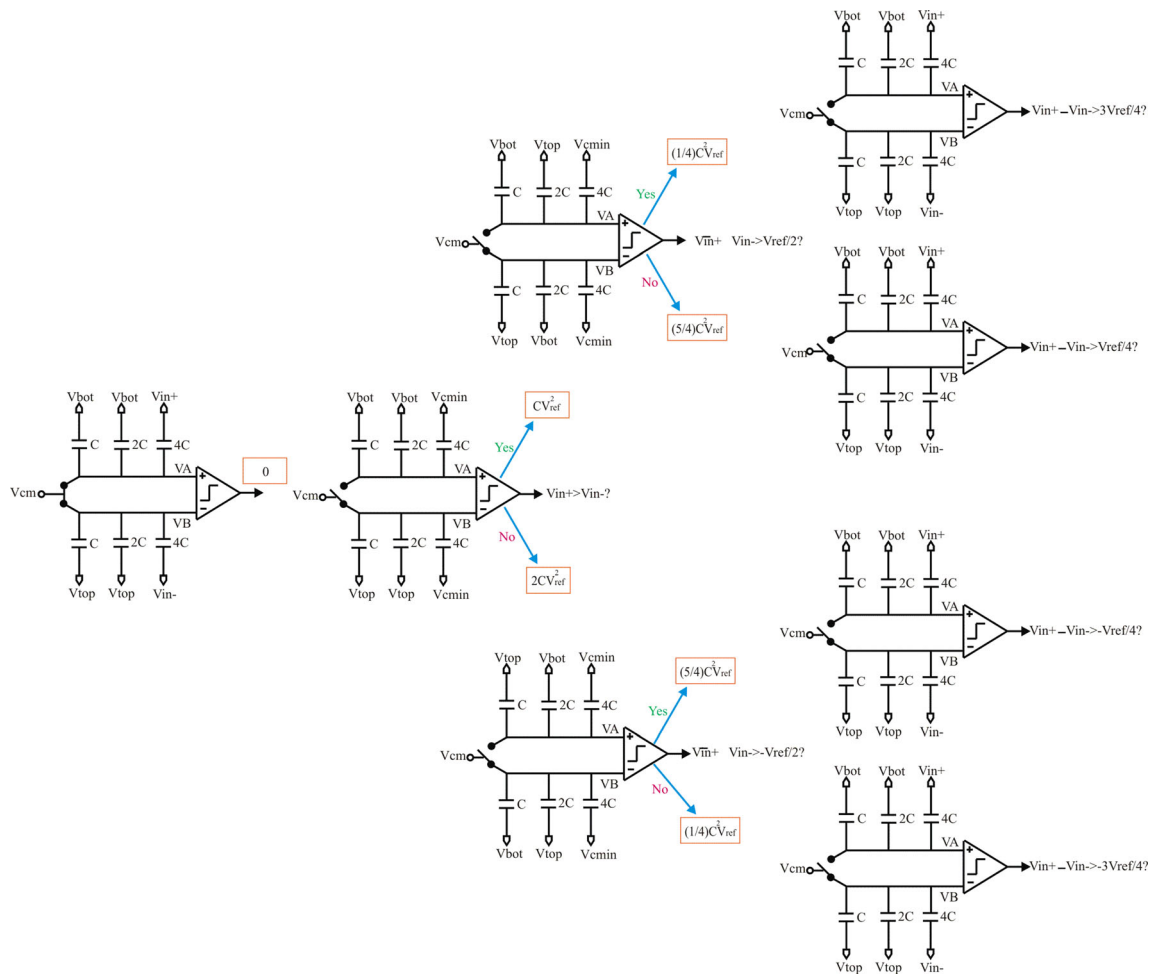


Fig. 3 Detail description of the proposed switching strategy

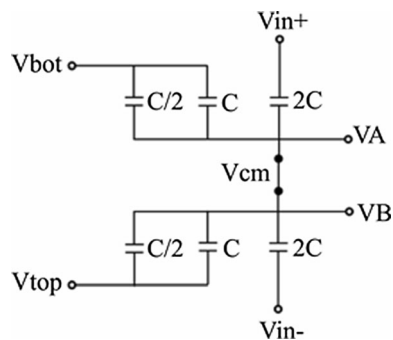


Fig. 4 The simplified circuit during the sampling mode for 3-bit example of the proposed SAR ADC

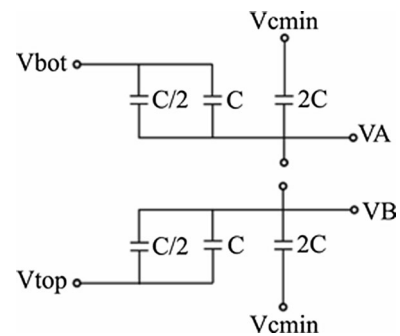


Fig. 5 The simplified circuit during the hold mode for 3-bit example of the proposed SAR ADC

bigger than VB, the comparator generates + 1 at output. Otherwise, the comparator provides − 1.

Figure 6 shows the simplified circuit for the next step when MSB = + 1. In these conditions the “2C” capacitance is connected to Vmin in the both sides, and “C” experiences Vtop and Vbot at the positive and negative sides of the differential structure, respectively. The rest of

the capacitors are remained unchanged. Calculations for the second bit extraction, based on the charge conservation law, are summarized in (5) and (6). In order to extract the third bit, the capacitors “2C” are connected to Vmin in the both sides, the capacitors “C” is switched to Vbot and Vtop, and the capacitors “C/2” are associated with Vtop and Vbot at the positive and negative sides of the ADC,

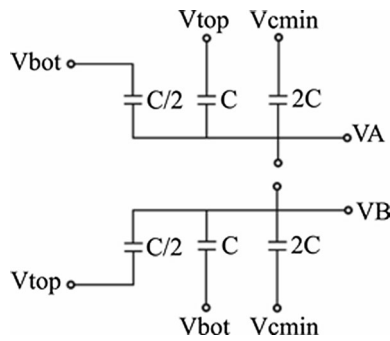


Fig. 6 The simplified circuit during the bit cycling phase for 3-bit example of the proposed SAR ADC

respectively. The simplified circuits are illustrated in Fig. 7 and related calculations are presented in (6) and (7). As the result, VB and VA are increased and decreased by the first binary weight, respectively, at the first cycle.

$$\begin{aligned} & [(V_{cmin} - V_A) - (V_{in+} - V_{cm})] \cdot 2C + [(V_{bot} - V_A) - (V_{bot} - V_{cm})] \cdot \left(1 + \frac{1}{2}\right)C \\ & = 0 \Rightarrow - \left(\underbrace{V_A - V_{cm}}_{\delta_{v+}} \right) \cdot \frac{7}{2}C - (V_{in+} - V_{cmin}) \cdot 2C = 0 \end{aligned} \quad (3)$$

$$\begin{aligned} \delta_{v+} &= - \left(\frac{4}{7} \right) \cdot (V_{in+} - V_{cmin}), \\ \delta_{v-} &= - \left(\frac{4}{7} \right) \cdot (V_{in-} - V_{cmin}) \Rightarrow \delta_v = \delta_{v+} - \delta_{v-} \\ &= - \left(\frac{4}{7} \right) \cdot (V_{in+} - V_{in-}) \Rightarrow \text{Bit1} = 1 \end{aligned} \quad (4)$$

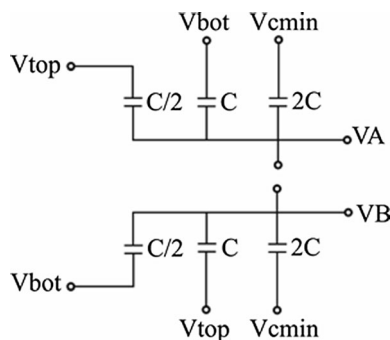


Fig. 7 The simplified circuit during the last bit extraction in 3-bit example of the proposed SAR ADC

$$\begin{aligned} & [(V_{cmin} - V_A) - (V_{in+} - V_{cm})] \cdot 2C \\ & + [(V_{bot} - V_A) - (V_{bot} - V_{cm})] \cdot 1C \\ & + [(V_{top} - V_A) - (V_{bot} - V_{cm})] \cdot \frac{1}{2}C = 0 \end{aligned} \quad (5)$$

$$\begin{aligned} & \Rightarrow - \left(\underbrace{V_A - V_{cm}}_{\delta_{v+}} \right) \cdot 7C - (V_{in+} - V_{cmin}) \cdot 4C \\ & - (V_{bot} - V_{top}) \cdot 2C = 0 \\ \delta_{v+} &= - \left(\frac{4}{7} \right) \cdot (V_{in+} - V_{cmin}) - \left(\frac{2}{7} \right) \cdot (V_{bot} - V_{top}), \\ \delta_{v-} &= - \left(\frac{4}{7} \right) \cdot (V_{in-} - V_{cmin}) - \left(\frac{2}{7} \right) \cdot (V_{top} - V_{bot}) \\ \Rightarrow \delta_v &= \delta_{v+} - \delta_{v-} = - \left(\frac{4}{7} \right) \cdot [(V_{in+} - V_{in-}) \\ & + (V_{bot} - V_{top})] \Rightarrow \text{Bit2} = 0 \end{aligned} \quad (6)$$

$$\begin{aligned} & [(V_{cmin} - V_A) - (V_{in+} - V_{cm})] \cdot 2C + [(V_{bot} - V_A) - (V_{bot} - V_{cm})] \cdot 1C + [(V_{top} - V_A) - (V_{bot} - V_{cm})] \\ & \cdot \frac{1}{2}C = 0 \Rightarrow - \left(\underbrace{V_A - V_{cm}}_{\delta_{v+}} \right) \cdot 7C - (V_{in+} - V_{cmin}) \\ & \cdot 4C - (V_{top} - V_{bot}) \cdot 2C = 0 \end{aligned} \quad (7)$$

$$\begin{aligned} \delta_{v+} &= - \left(\frac{4}{7} \right) \cdot (V_{in+} - V_{cmin}) - \left(\frac{4}{7} \right) \cdot (V_{bot} - V_{top}), \\ \delta_{v-} &= - \left(\frac{4}{7} \right) \cdot (V_{in-} - V_{cmin}) - \left(\frac{4}{7} \right) \cdot (V_{top} - V_{bot}) \\ \Rightarrow \delta_v &= \delta_{v+} - \delta_{v-} = - \left(\frac{4}{7} \right) \cdot (V_{in+} - V_{in-}) \\ & - (2/7)(V_{bot} - V_{top}) \Rightarrow \text{Bit3} = 1 \end{aligned} \quad (8)$$

As a brief rule for expanding to more bits, the following steps can be organized:

- At the first step, the sign bit is extracted by connecting the bottom plate of the MSB capacitor from V_{in} to the V_{cmin} potential, while the top plate is disconnected from V_{cm} .
- For the next bit extraction, if the last extracted bit is “1”, then the reference voltages are swapped in only the capacitors of the next bit. Else, in conditions that the last extracted bit is “0” then the reference voltages are swapped in both the capacitors of the current and the next bits.

3 Circuit implementation

3.1 Digital to analog converter (DAC)

Producing the analog voltage based on the results of the previous bits is devolved to DAC as one of the most significant sub-blocks in SAR ADCs [13, 18, 26, 35]. Basically, for a high-resolution SAR ADC, highly linearized capacitor DAC array is also required, [10, 20, 31]. It is noticeable that, for each additional bit, number of the capacitors would considerably grow up and the area of the capacitor array would severely increase, consequently. Also, the increased switching power, capacitor switching time and input capacitive loading are expected, [1, 7, 9, 15, 17, 25, 39]. In order to resolve these drawbacks a novel CDAC structure is presented based on the well-known split technique, as illustrated in Fig. 8. As discussed above in the split CDAC structure, size of the coupling capacitor is one of the critical issues, because dominantly decides the accuracy of the CDAC. On the other hand, in the layout design, mismatch between capacitors should be cautiously considered, [1]. It is noteworthy that, in the layout design all the capacitors are made based on the unit capacitor. Hence, the coefficient of the coupling capacitor (C_c) is selected as the same as the unit one, used for LSB, to improve the linearity, remove the mismatch nonidealities and tolerate the process gradients. Consider that the value of the unit capacitor ($C/16$) is selected 62.5fF. Likewise, the common-centroid layout technique is utilized to achieve further matching results. Meanwhile, to ensure that all unit capacitors in the capacitor array are subject to identical capacitive loads a dummy capacitor ring is utilized on capacitor array.

Again the 3-bit example is surveyed in Figs. 9, 10, 11, 12, 13, 14 and 15 to clarify the proposed idea. As it is clear in Fig. 10, if MSB bits are obtained “1”, then the “4C” capacitor is connected to V_{ref} , while the others are

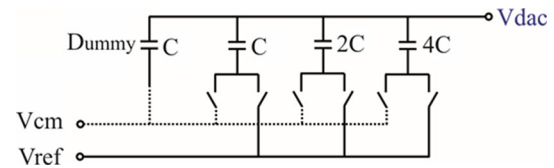


Fig. 9 A 3-bit capacitor DAC

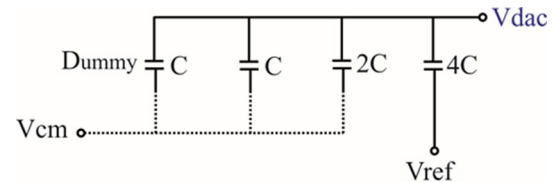


Fig. 10 A 3-bit capacitor DAC with first MSB bit is “ON”

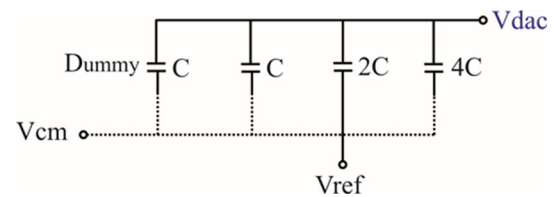


Fig. 11 A 3-bit capacitor DAC with second MSB bit is “ON”

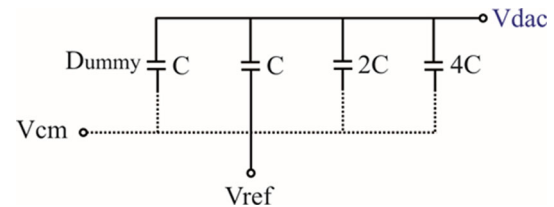


Fig. 12 A 3-bit capacitor DAC with LSB bit is “ON”

associated to V_{cm} . In these conditions, the output voltage of the 3-bit DAC can be calculated as (9):

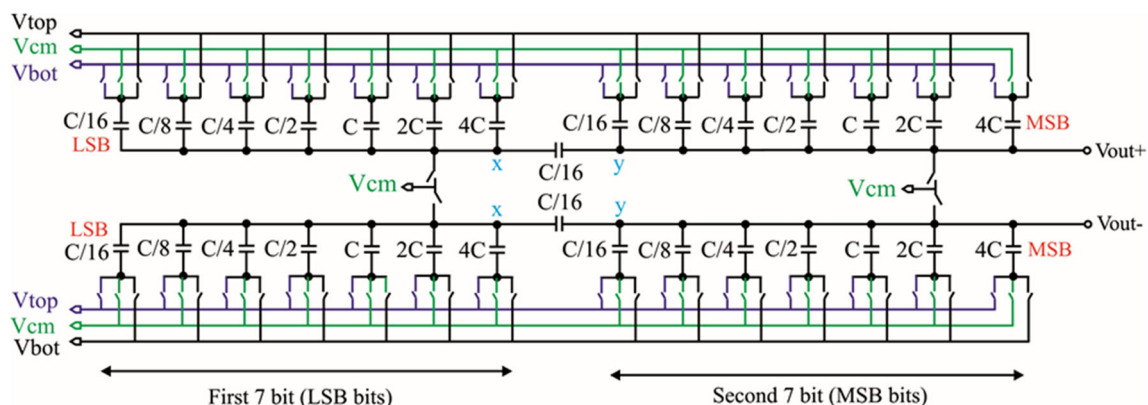


Fig. 8 The proposed fully differential capacitor DAC

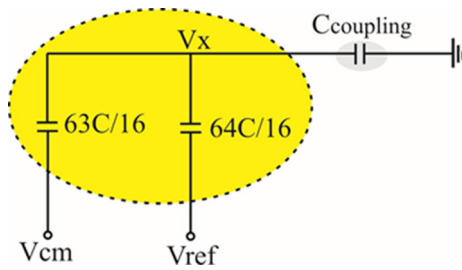


Fig. 13 A small case of the proposed 14-bit capacitor DAC with MSB bit of the first seven bits is “ON”

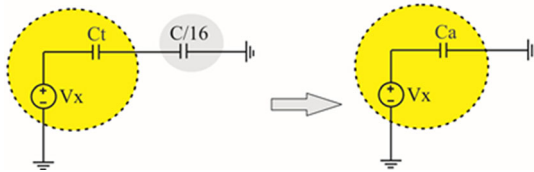


Fig. 14 The Thevenin model of the circuit

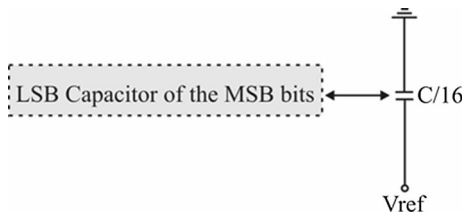


Fig. 15 A 14-bit capacitor DAC with LSB bit of the second seven bits (MSB bits) is “ON”

$$V_{dac} = \frac{4C \cdot V_{ref}}{C + C + 2C + 4C} = \left(\frac{1}{2}\right) V_{ref} \quad (9)$$

it is noticeable that, for simplicity of the mathematical calculations, V_{cm} is assumed to 0 V in this example. In addition, if the second and the third bits are “1”, as shown in Figs. 11 and 12, then the values of the output voltages of DAC are retrieved according to (10) and (11):

$$V_{dac} = \frac{2C \cdot V_{ref}}{C + C + 2C + 4C} = \left(\frac{1}{4}\right) V_{ref} \quad (10)$$

$$V_{dac} = \frac{C \cdot V_{ref}}{C + C + 2C + 4C} = \left(\frac{1}{8}\right) V_{ref} \quad (11)$$

(9), (10) and (11) prove that the injected charge, initiated from the third bit, is half of the second one, and the injected charge due to the second bits is half of the first one. According to Fig. 13, the MSB (relates to $64C/16$ capacitor) from the first seven bits (LSB section) is supposed to be “1” and the other bits are assumed to be “0”, which are connected to V_{ref} and V_{cm} respectively. Meanwhile, the potential of the x node (V_x) is calculated in (12) which helps to achieve the value of the injected charge in the x node (Q_x):

$$V_x = \left[\frac{\left(\frac{64C}{16}\right) \cdot V_{ref}}{\left(\frac{63C}{16}\right) + \left(\frac{64C}{16}\right)} \right] = \left(\frac{64}{127}\right) \cdot V_{ref} \quad (12)$$

The Thevenin model of Fig. 13 is presented in Fig. 14, which can be used to simplify calculating the C_c . For this case, the value of the injected charge in the x node (Q_x) can be computed from (13) to (15):

$$C_t = \left[\left(\frac{63C}{16}\right) + \left(\frac{64C}{16}\right) \right] = \left(\frac{127}{16}\right) C \quad (13)$$

$$C_a = \left[\left(C_t \cdot \left(\frac{C}{16}\right) \right) / \left(C_t + \left(\frac{C}{16}\right) \right) \right] = \left(\frac{127}{128}\right) \cdot \left(\frac{1}{16}\right) C \quad (14)$$

$$Q_x = (V_x \cdot C_a) = \left[\left(\frac{64}{127}\right) V_{ref} \right] \left[\left(\frac{127}{128}\right) \left(\frac{1}{16}\right) C \right] = \left(\frac{1}{2}\right) \left(\frac{1}{16}\right) (V_{ref} \cdot C) \quad (15)$$

On the other hand, if the LSB bit (relates to $C/16$ capacitor) from the second seven bits (MSB bits) is supposed to be “1”, as clarified in Fig. 15. In these conditions the value of the injected charge in y node (Q_y) can be calculated from (16) to (19):

$$V_y = V_{ref} \quad (16)$$

$$C_y = \left(\frac{1}{16}\right) C \quad (17)$$

$$Q_y = (V_y \cdot C_y) = \left(V_{ref} \cdot \left(\frac{1}{16}\right) C \right) = \left(\frac{1}{16}\right) \cdot (V_{ref} \cdot C) \quad (18)$$

$$Q_x = \left(\frac{1}{2}\right) Q_y \quad (19)$$

Consistently, as it is clear in (19) the value of Q_x is half of the Q_y . It means that if the size of the coupling and unit capacitor are chosen the same, then the injected charge in the seventh bit is half of the eighth bit. As the result, applying this idea the coefficient of the coupling capacitor can be eliminated reliably, and the injected charge in the previous bit is half of the next bit and so on.

In the CDAC-based SAR ADC another important feature is how to connect the bottom or top plate of the capacitors to minimize the destroying effect of the parasitic

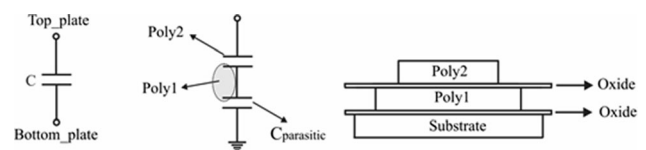


Fig. 16 Implementing poly-poly capacitor in CMOS process

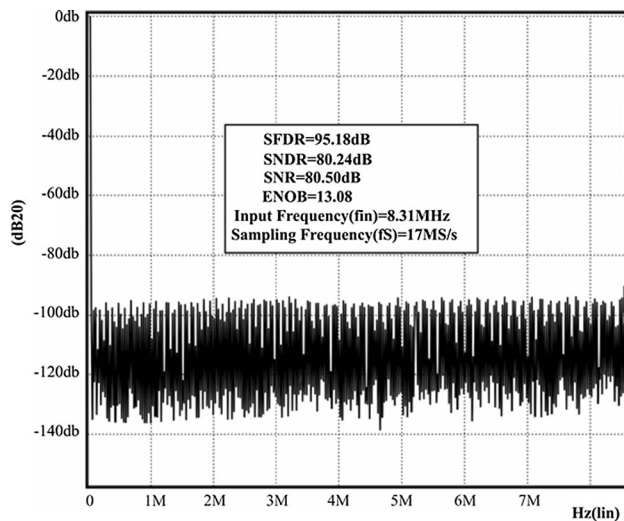


Fig. 18 The output FFT spectrum of the proposed ADC at $f_S = 17$ MS/s and near the Nyquist input frequency of 8.31 MHz

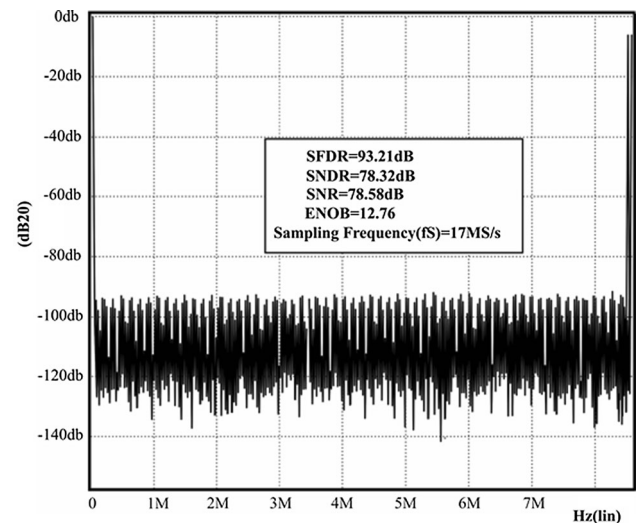


Fig. 20 Double tone output FFT spectrum of the proposed ADC in $fin1 = 8.31$ MHz and $fin2 = 8.34$ MHz at $f_S = 17$ MS/s

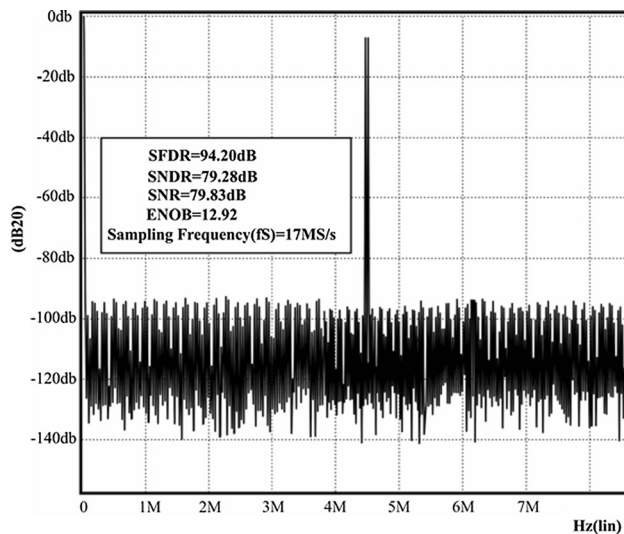


Fig. 19 Double tone output FFT spectrum of the proposed ADC at $f_S = 17$ MS/s for $fin1 = 4.15$ MHz and $fin2 = 4.18$ MHz

input (differential peak-to-peak voltage is 4 V). Results confirm that the SNDR and SFDR of 80 dB and 95 dB are achieved, respectively, and the Effective Number of Bits (ENOB) is reached to around 13.08-bits. Meanwhile, double-tone simulation results with input frequency of 4.15 MHz and 4.18 MHz, 8.31 MHz and 8.34 MHz at 17 MS/s sampling rate with the differential peak-to-peak range of 4 V are depicted in Figs. 19 and 20, correspondingly.

As shown in Figs. 21 and 22, INL and DNL of the proposed ADC experience the maximum of 0.95/− 0.58 LSB and + 0.63/− 0.74 LSB, respectively.

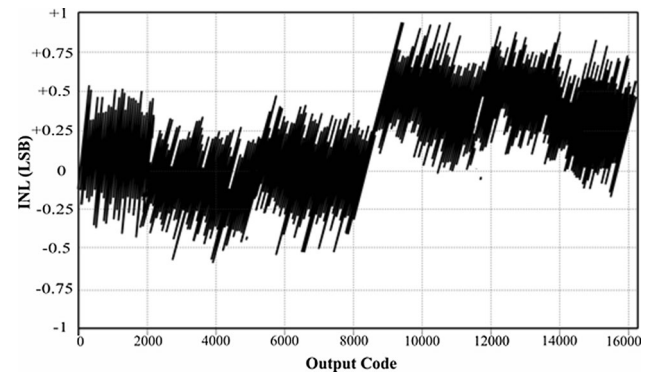


Fig. 21 Simulation result of INL plot

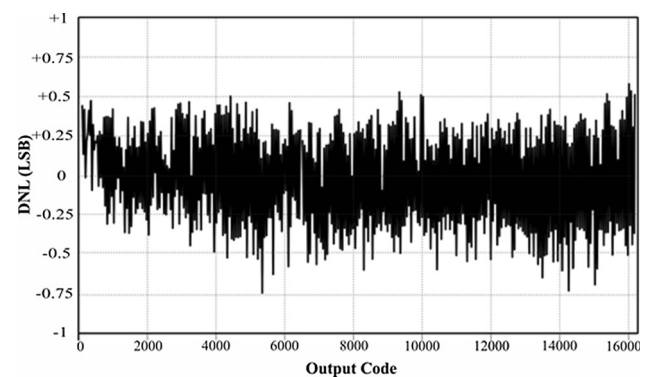


Fig. 22 Simulation result of DNL plot

Figure 23 summarizes the simulated SNDR and SFDR versus the input frequency at $f_S = 17$ MS/s sampling rate for all process corners (TT, FS, SF, SS, and FF). Consider

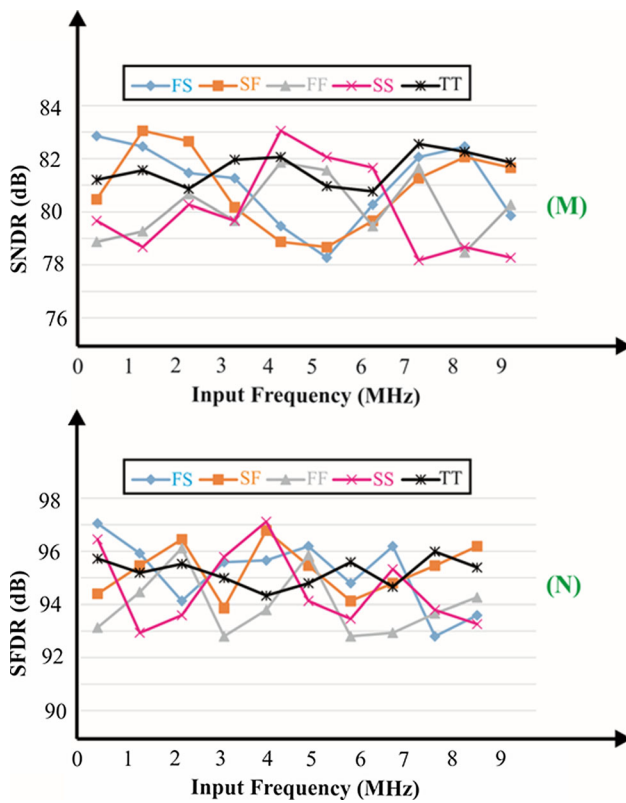


Fig. 23 Post-layout simulation results of SNDR (M) and SFDR (N) versus the input frequency at $f_s = 17$ MS/s, for all process corners (TT, FS, SF, SS and FF)

that the SNDR and SFDR are labeled by (M) and (N), respectively. It is concluded from Fig. 23 that at the sampling rate of 17 MS/s and the Nyquist input rate of 8.31 MHz, the SNDR and SFDR are retrieved above 78 dB and 93 dB, respectively, for all corner conditions. The Walden and Schreier Figure-of-Merit (FOM) equations are used as (20) and (21), respectively:

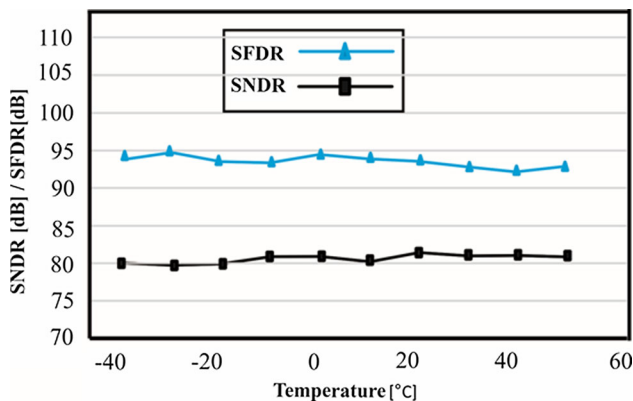


Fig. 24 Post-layout simulation results of SNDR and SFDR versus temperature with a nyquist input (8.31 MHz) at $f_s = 17$ MS/s

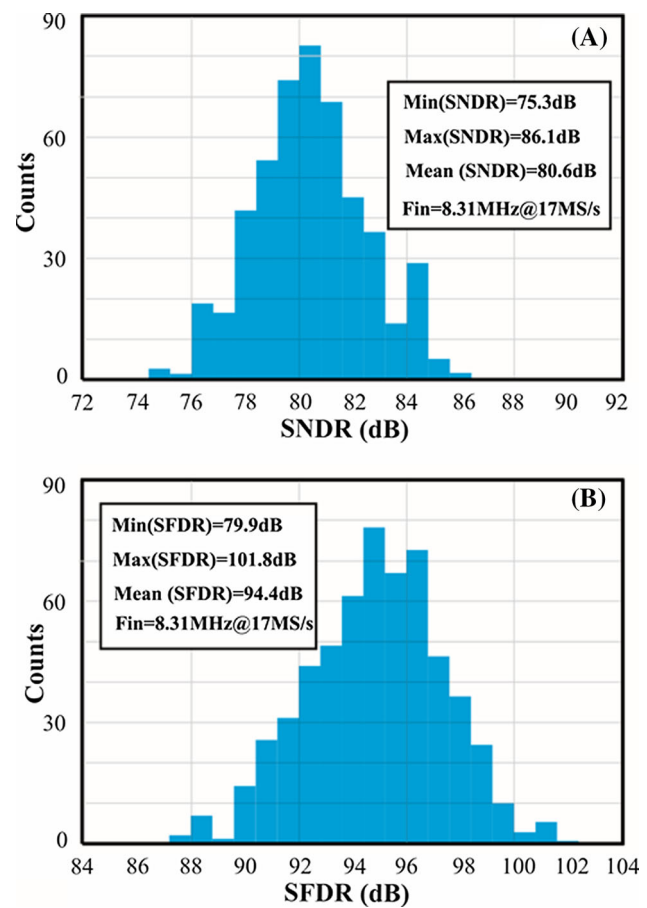


Fig. 25 Monte-Carlo analysis with 250 iterations by applying 1% mismatch on unit capacitors and 300 mV variations on reference voltage. (A) SNDR and (B) SFDR, at $f_s = 17$ MS/s and Nyquist input

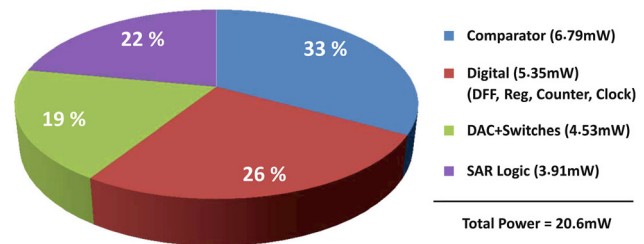


Fig. 26 Power breakdown

$$FOM_W = \frac{P_{total}}{2^{ENOB} * f_s} \quad (20)$$

$$FOM_S = SNDR + 10 \log \left(\frac{f_s}{2 * power} \right) \quad (21)$$

where P_{total} , $ENOB$, and f_s represent the total power consumption, the effective number of bits and the Nyquist sampling frequency, respectively. Calculated FOM_W and

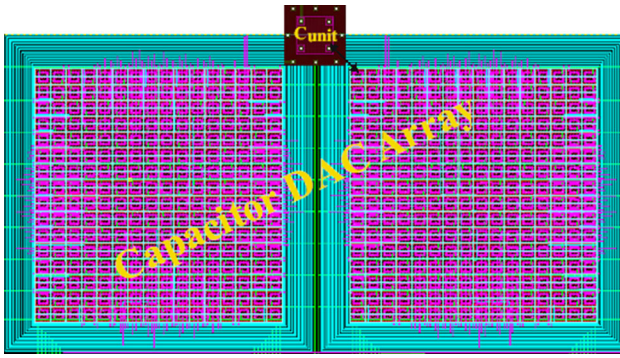


Fig. 27 Layout floorplan of the proposed capacitor DAC array

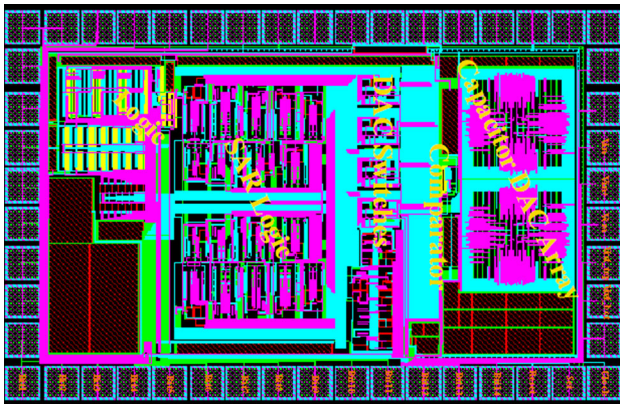


Fig. 28 Layout floorplan of the proposed SAR ADC

FOM_S based on (20) and (21) experience 139.9fJ and 166.15 dB per conversion step, correspondingly.

Table 2 Comparison table

Specifications	[2] ^a	[23] ^a	[3] ^a	[13] ^b	[19] ^a	This work ^b
Year	2015	2019	2018	2014	2015	2019
Technology (nm)	40	40	130	350	600	500
Resolution (bits)	14	12	14	14	14	14
Sampling rate (MS/s)	35	100	0.1	0.25	0.4	17
Supply voltage (V)	1.2/2.5	0.9	3.3/1.2	3.3	± 15	5
Total power (mW)	54.5	2.6	0.059	4.29	90	20.6
FOMW (fJ/Conv-Step)	—	14.6	45.42	2050	—	139.9
FOMS at Nyq. (dB)	159.5	—	—	—	—	166.15
SNDR at Nyq. (dB)	74.4	67.3	—	80.2	73.32	80
SFDR at Nyq. (dB)	90	81.8	94.28	97.8	—	95
ENOB at Nyq. (bit)	—	10.8	13.65	—	11.5	13.08
Power supply noise (mV)	—	—	—	—	—	100
INL (LSB)	0.67/− 0.86	0.68/− 0.93	+ 1.27/− 0.54	01.81/− 1.81	0.95	0.95/− 0.58
DNL (LSB)	0.73/− 0.43	1.13/− 0.96	+ 0.74/− 0.49	+ 0.55/− 0.55	0.92	0.63/− 0.74
Active area (mm ²)	0.236	0.014	0.934	0.98	9.76	1.7

^aFabrication

^bSimulation

Table 1 Size of transistors in comparator

Transistors	Aspect ratio (W/L)
M1, M2	20/0.50
M3, M4	30/0.55
M5, M6	30/0.55
M7, M8	10/0.50
M9, M10	20/0.50
M11, M12	60/0.55
M13, M14	20/0.55
M0, M01	40/0.50

The temperature dependency of SNDR and SFDR is summarized in Fig. 24, for -40°C to 60°C temperature range. It can be observed that less than 1.5 dB and 3 dB drop might be occur in SNDR and SFDR, for whole range of temperature variations, respectively.

The results of the Monte-Carlo analysis for 250 iterations, by applying mismatch on capacitors and intentional noise on comparators and the reference levels, is accomplished and the results are summarized in Fig. 25. In this simulation all the parasitic capacitances are considered and the capacitor mismatch is $\sigma_{\text{mismatch}} = 1\%$. Also, both reference noise and comparator noise error is assumed $\sigma_{\text{noise}} = 0.3\text{ V}$. As it is clear in Fig. 25a and b, the SNDR and SFDR of the proposed ADC, experience the worst/best values of about 75.3 dB/86.1 dB and 79.9 dB/101.8 dB during 250 iterations. Standard deviations throughout all

iterations reaches to 1.9 dB and 1.6 dB for SNDR and SFDR, respectively.

Separated power consumption for different blocks are depicted in pie diagram of Fig. 26. As shown, total power consumption is 20.6 mW at 5 V power supply. Comparator block participates with more power consumption, means around 6.79 mW.

Layout floorplan of the CDAC array and the proposed 14-bit SAR ADC is indicated in Figs. 27 and 28, respectively. It has been designed in 0.5 μm standard CMOS technology and the active area is around 1.7 mm² (1.7 mm $\times$ 1 mm). Size of transistors of the comparators devices are summarized in Table 1.

5 Conclusions

This paper presents a new fully differential SAR ADC structure with improved switching strategy and enhanced linearity. A novel technique is presented to decrease the total capacitance and total switches in CDAC structure besides increasing the speed of the ADC. It is noteworthy that, applying the suggested method the total switches are reduced by about 55% rather than the conventional strategies. Moreover, in order to improve the linearity, remove the mismatch effects and tolerates the process gradients of the capacitor DAC, the coefficient of the coupling capacitor (C_c) is removed in the split capacitor technique as well, it means that the size of the coupling and LSB capacitors are chosen the same. Meanwhile, at the 5 V supply voltage and 17 MS/s sampling rate at Nyquist input rate ($f_{in} = 8.31$ MHz) and with the differential peak-to-peak voltage of 4 V, the ADC experiences SNDR and SFDR of 80 dB and 95 dB, respectively. Effective number of bits (ENOB) of 13.08 bit is also achieved. Total power consumption of the proposed ADC is 20.6 mW at 5 V supply potential. Table 2 compares the specifications of the proposed SAR ADC with the other similar works.

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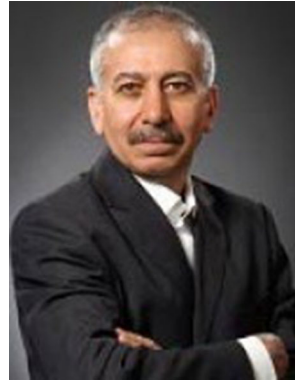
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